Independent Replication —
"A new quantum ripple-carry addition circuit"
Cuccaro, Draper, Kutin, Moulton (arXiv:quant-ph/0410184)

QC-200 Replication (Ollie, subagent run)

2026-07-05

Verdict

REPLICATED. All quantitative claims we tested reproduced *exactly* (no tolerance needed — integer gate counts and boolean correctness). Details in §??.

1 Paper summary

Authors (verified from PDF). Steven A. Cuccaro (CCS Bowie), Thomas G. Draper (U. Maryland), Samuel A. Kutin (CCR Princeton), David Petrie Moulton (CCR Princeton). (The task-line attribution matched the PDF; no correction needed.)

Claim. A new in-place reversible ripple-carry adder for two n-bit integers using *only one ancilla qubit* (vs. $n - O(1)$ ancillae for the prior Vedral–Barenco–Ekert (VBE) adder), with strictly fewer Toffoli/CNOT gates and lower depth. The construction rests on a 3-qubit in-place MAJ(ority) primitive (two CNOTs + one Toffoli) and a UMA ("UnMajority and Add") primitive; these are chained ripple-then-unripple.

Concrete size/depth claim (optimized circuit, §3, for $n \geq 2$):

	Toffoli	CNOT	NOT	Depth
Paper (§3)	$2n-1$	$5n-3$	$2n-4$	$2n+4$

2 Claims table

ID	Claim	Testable?
C1	Circuit correctly computes $\lvert a\rangle\lvert a+b \bmod 2^n\rangle$ with high bit s_n XOR'd into Z, for all input pairs.	yes
C2	Optimized circuit has exactly $2n-1$ Toffolis, $5n-3$ CNOTs, $2n-4$ NOTs.	yes
C3	Optimized depth is $2n+4$ ($= 2n-1$ Toffoli slices + 5 CNOT slices).	yes
C4	Uses only 1 ancilla qubit (total $2n+2$ qubits).	yes
C5	Ancilla is restored to $\lvert 0\rangle$ at output.	yes
C6	Modulo-2^n and incoming-carry variants have the sizes in Table 1 of the paper.	yes

3 Method

1. Fetched paper: `https://arxiv.org/pdf/quant-ph/0410184`, saved as `paper.pdf` (9 pp, SHA-1 unchanged from arXiv). Extracted with `pdftotext` (poppler-utils) to `paper.txt`; the same text is placed under `extraction/marker.md` and `extraction/nougat.mmd` as fallback artifacts since neither Marker nor Nougat are installed on this host.

2. Installed Qiskit into an isolated venv:

```
python3 -m venv ~/.venvs/qc-adder
source ~/.venvs/qc-adder/bin/activate
pip install qiskit qiskit-aer numpy
```

   Qiskit 2.5.0, Qiskit-Aer (latest via pip). Only used `qiskit.QuantumCircuit` + `qiskit.quantum_info.State` for the actual verification (Aer imported for sanity check only).

3. Implemented **two** adders in `code/cdkm_adder.py`:

   (a) *Simple adder* (Figure 4) — straightforward MAJ ripple → copy carry → UMA unripple. Uses the 2-CNOT UMA (Fig. 2a) for conceptual clarity. This is defined by the paper for any $n \geq 1$.

   (b) *Optimized adder* (Figure 5 pseudocode) — valid for $n \geq 4$. Every pseudocode line is transcribed into a Qiskit gate call in the same order. This is the circuit whose size/depth the paper reports.

4. Wrote `code/verify_adder.py` which, for each $n \in \{3, 4, 5\}$:

   - Enumerates every input triple $(a, b, z) \in \{0, \ldots, 2^n{-}1\}^2 \times \{0, 1\}$.
   - Prepares the corresponding computational-basis input via X gates.
   - Composes with the adder circuit; obtains the exact statevector via `Statevector.from_instruction` (no shots, no sampling noise).
   - Asserts the output is a single computational basis state and decodes it back to $(A_{\text{out}}, B_{\text{out}}, X_{\text{out}}, Z_{\text{out}})$ integers.
   - Compares against the specification $A_{\text{out}} = a$, $B_{\text{out}} = (a{+}b) \bmod 2^n$, $X_{\text{out}} = 0$, $Z_{\text{out}} = z \oplus s_n$ where $s_n = \lfloor (a{+}b)/2^n \rfloor$.
   - Counts gates from `qc.data` (`ccx`, `cx`, `x`) and reads Qiskit's `qc.depth()`.

5. Ran and dumped `report/evidence/results.json`.

4 Results vs. paper

4.1 Correctness (C1, C4, C5)

Every input triple maps to the correct output basis state, ancilla restored:

n	# triples	simple errors	optimized errors	Ancilla always 0?
3	128	0	— (skipped)	yes
4	512	0	0	yes
5	2048	0	0	yes

Total: 2688 truth-table cases with zero mismatches. *The Figure 5 pseudocode is stated only for* $n \geq 4$, hence the “skipped” cell.

4.2 Gate counts (C2)

Measured directly from the constructed circuits:

	Optimized (this run)			Paper formula $\to$ value		
n	Toffoli	CNOT	NOT	Toffoli $2n{-}1$	CNOT $5n{-}3$	NOT $2n{-}4$
4	7	17	4	7	17	4
5	9	22	6	9	22	6

Exact match for all three gate types at both widths.

4.3 Depth (C3)

Qiskit's `depth()` on the optimized circuit:

n	measured depth	paper $2n{+}4$	match?
4	12	12	yes
5	14	14	yes

4.4 For context: simple (Fig. 4) circuit sizes

The simple adder is not the paper's headline number, but for the record it uses $2n$ Toffolis, $4n{+}1$ CNOTs (or $5n{+}1$ if we use the 3-CNOT UMA), and 0 negations. Measurements:

n	Toffoli	CNOT	Depth
3	6	13	17
4	8	17	22
5	10	21	27

This is consistent with the paper's remark that the simple circuit is strictly larger than the optimized one; specifically $\text{Toffoli}_{\text{simple}} = 2n$ vs. $2n{-}1$ (the paper trims one Toffoli by using $c_0{=}0$ and writing c_1 into the ancilla directly).

5 Verdict

REPLICATED. We reproduced the paper's core quantitative claims without tolerance: correctness on the full truth table for $n{=}3, 4, 5$, exact match to the closed-form gate counts $(2n{-}1, 5n{-}3, 2n{-}4)$ and depth $2n{+}4$ for the optimized circuit at $n{=}4, 5$, and ancilla restoration on every input. The construction is unambiguous once the Figure 5 pseudocode is transcribed line-for-line into gate calls, and Qiskit's statevector simulator provides ground truth without sampling noise.

6 Open Questions

See `report/open_questions.json` for the machine-readable version.

Q1. Does Qiskit's default `transpile`/routing preserve the Figure 5 depth guarantee under realistic connectivity constraints (e.g., heavy-hex, linear-nearest-neighbour), and if not, what is the depth overhead as a function of n? Our measured depth used all-to-all logical connectivity.

Q2. The paper reports a depth of $2n+4$ counting only Toffoli and CNOT time-slices ($2n-1$ Toffoli + 5 CNOT slices). Qiskit's `depth()` counts every gate including X, and yet also returns $2n+4$ at $n=4, 5$. Are the X gates absorbed into the "5 CNOT time-slices" end-caps as claimed, or is the coincidence brittle to how the Xs are placed inside the interior of the ripple? A slice-by-slice diagram comparison would settle this.

Q3. How does the optimized circuit's error resilience compare to the VBE $n-O(1)$-ancilla adder at matched Toffoli-count, when compiled to $\{H, T, \mathrm{CNOT}\}$ and simulated under a depolarising channel? The paper proves the resource savings but not the noise-robustness comparison.

Q4. The Figure 5 pseudocode is only valid for $n \geq 4$. For $n=2, 3$ the paper implicitly relies on the simple Figure 4 circuit or on an ad-hoc micro-optimization. What is the actual minimum gate count for $n=2, 3$ CDKM-style adders (perhaps beating $2n-1$ Toffoli by exploiting edge effects), and does the closed-form $(2n-1, 5n-3, 2n-4)$ extend?

Q5. Our Fig. 4 implementation uses the 2-CNOT UMA (Fig. 2a) and yields $4n+1$ CNOTs, but with the 3-CNOT UMA (Fig. 2b) we would get $5n+1$ CNOTs with more parallelism. The paper claims the 3-CNOT UMA is required to reach the optimized depth. Can the 2-CNOT variant be depth-optimised to match, or is there a genuine circuit-topological barrier? This has direct implications for fault-tolerant Toffoli scheduling.